The Martingale Overlap Estimate for MPS Parent Hamiltonians

2026-06-04

1 The source assertion

The martingale-method paragraph in Cirac–Perez-Garcia–Schuch–Verstraete 2021 can be compared with the quantitative estimate isolated in the formal parent-Hamiltonian proof.[1] The relevant source passage is the discussion of gaps for MPS parent Hamiltonians in [CPGSV21, Section IV.C], where the local paragraph `sec:4:hams-gaps` develops the inequalities labelled `eq:4:martingale-1` and `eq:4:martingale-2`.

Let

$$H = \sum_i h_i$$

be a frustration-free Hamiltonian whose local terms are projections. The source recalls that a gap lower bound $\gamma > 0$ is equivalent, at the quadratic-form level, to

$$H^2 \geq \gamma H.$$

Since $h_i^2 = h_i$, the expansion of H^2 separates diagonal terms, overlapping products, and non-overlapping products. The non-overlapping products are treated as non-negative. Thus it is enough to control each overlapping pair by an anticommutator estimate

$$h_i h_j + h_j h_i \geq -c_{ij}(1-\gamma)(h_i + h_j), \qquad (\texttt{eq:4:martingale-2})$$

where the coefficients c_{ij} are chosen so that the contributions in each row add up to one. The formal row-sum algebra uses the slightly more flexible condition that, for each fixed i, the sum of the coefficients attached to terms overlapping h_i is at most one.

The same paragraph interprets this estimate as a lower bound on the smallest non-zero principal angle between the local ground spaces of overlapping terms. It then cites the martingale method and the MPS parent-Hamiltonian results of Fannes–Nachtergaele–Werner and Nachtergaele, together with the later

[1] For traceability, this note corresponds to issue #1044. The remaining quantitative estimate is recorded in issue #952, and the broader parent-Hamiltonian theorem in issue #460.

Kastoryano–Lucia formulation, for the conclusion that MPS parent Hamiltonians have a uniform positive spectral gap after a suitable blocking [FNW92, Nac96, KL18]. The review text does not spell out the numerical constants or the precise finite cyclic blocking convention needed for the formal statement below.

2 The formal reduction

The formal development has separated the finite-row martingale algebra from the remaining MPS-specific principal-angle estimate. Fix an interaction range $L > 1$ and an N-site periodic chain with $N \geq 2L$. Let p_i denote the transported length-L local parent-Hamiltonian projection at the cyclic window starting at i. Write $i \sim_L j$ when the two length-L cyclic windows intersect and $i \neq j$. For each fixed i, the number of indices j with $i \sim_L j$ is at most

$$m = 2(L-1).$$

The formal row-sum reduction specializes the source coefficients to

$$c_{ij} = \frac{1}{2(L-1)}$$

for overlapping cyclic windows and to 0 otherwise. The row-cardinality bound then gives $\sum_j c_{ij} \leq 1$.

With this choice, the constant in the formal statement is

$$\gamma_L = \frac{1}{4L}.$$

The ordered overlapping estimate sufficient for the gap bound is

$$\mathrm{Re}\langle p_i v, p_j v\rangle \geq -\left(1 - \frac{1}{4L}\right)\frac{1}{2(L-1)}\,\mathrm{Re}\langle p_i v, v\rangle \tag{F}$$

for every $N \geq 2L$, every off-diagonal overlapping pair $i \sim_L j$, and every vector v in the finite chain Hilbert space. Given this estimate, the already formalized row-sum argument proves $H_{N,L}^2 \geq \gamma_L H_{N,L}$, and hence a norm lower bound with constant γ_L on the orthogonal complement of the ground space.

The formal proof also isolates a sufficient norm-compression form of the same input:

$$\|p_i p_j v\| \leq \left(1 - \frac{1}{4L}\right)\frac{1}{2(L-1)}\|p_i v\|. \tag{N}$$

For a symmetric projection p_i one has

$$\mathrm{Re}\langle p_i v, p_j v\rangle = \mathrm{Re}\langle p_i v, p_i p_j v\rangle \geq -\|p_i v\|\|p_i p_j v\|.$$

Thus (N), together with $\mathrm{Re}\langle p_i v, v\rangle = \|p_i v\|^2$, implies (F). This is the projection-geometry comparison used in the formal proof: a norm-compression estimate for each overlapping pair is converted by Cauchy–Schwarz to the ordered Friedrichs estimate, and the finite row-sum argument converts those ordered estimates into the global gap bound.[2]

3 What remains to compare

The source discussion and the formal theorem are aligned at the qualitative level: both use the martingale expansion, non-negativity of non-overlapping terms, row-sum coefficients for overlapping terms, and principal-angle estimates for local ground spaces. The point still requiring a mathematical check is quantitative and conventional.

First, the formal theorem fixes the uniform coefficient $1/(2(L-1))$ for every overlapping cyclic-window pair and asks for the explicit constant $\gamma_L = 1/(4L)$. The review text states the general martingale condition with coefficients c_{ij} whose rows add to one, and it refers to blocking and principal angles to obtain a positive gap. It does not by itself show that the Fannes–Nachtergaele–Werner, Nachtergaele, or Kastoryano–Lucia estimates provide exactly the bound (F), or the stronger sufficient bound (N), for this fixed cyclic-window convention and for this particular numerical value of γ_L.

Second, the blocking parameter in the cited martingale argument must be matched with the formal convention. The formal statement is uniform for all $N \geq 2L$ and for the length-L cyclic windows used in the formal finite-chain Hilbert-space realization. A source proof that produces some positive gap after increasing or changing the blocking may still be mathematically sufficient for the existential MPS gap theorem, but it would not immediately be the displayed all-L estimate with constant $1/(4L)$ unless the constants are transported through the same blocking convention.

Consequently the remaining comparison is not whether the row-sum algebra is valid; that part has been isolated and formalized. The remaining comparison is whether the cited FNW–Nachtergaele–Kastoryano principal-angle estimates supply the precise overlapping cyclic-window estimate required by (F) or (N), or whether the formal development is using a stronger replacement input than the review text states explicitly.

4 Relation to the blueprint

The blueprint item labelled `lem:martingale_mps` in `blueprint/src/chapter/ch14_parent_hamiltonian.tex` now separates the finite-row martingale

[2] The current formal anchors are the cyclic-window reductions parentHamiltonianES_gap_bound_of_cyclic_window_overlap_friedrichs, parentHamiltonianES_gap_bound_of_cyclic_window_overlap_norm_bound, and parentHamiltonianES_gap_bound_of_cyclic_window_overlap_norm_bound_of_le in `TNLean/MPS/ParentHamiltonian/Martingale/Reduction.lean`. The remaining theorem is parentHamiltonianES_gap_bound_of_friedrichs in `TNLean/MPS/ParentHamiltonian/Martingale/Gap.lean`. The projection-geometry reductions used by these statements are in `TNLean/Analysis/ProjectionGeometry.lean`.

reduction from the still-open quantitative estimate. It does not assert that the displayed all-$L > 1$ cyclic-window bound has already been obtained from the cited principal-angle estimates. Instead, it identifies Theorem `thm:friedrichs_gap_bound` as the remaining analytic input.

The corresponding theorem entry `thm:friedrichs_gap_bound` points to parentHamiltonianES˙gap˙bound˙of˙friedrichs in `TNLean/MPS/ParentHamiltonian/Martingale/Gap.lean`. This is the current formal boundary: all finite-row, cyclic-window, non-overlap, and norm-compression reductions have been isolated in `TNLean/MPS/ParentHamiltonian/Martingale/Reduction.lean`, while the MPS-specific Friedrichs estimate remains to be proved or matched precisely to the cited martingale estimates.

Once the quantitative comparison with the cited martingale estimates is settled, the blueprint should say either that the all-L bound follows from those estimates under the cyclic-window convention used here, or that the formal proof uses a deliberately stronger replacement estimate.

5 Conclusion

The conclusion is a difference between the review paragraph and the formal quantitative statement, with one mathematical comparison still to be made. The source states the martingale condition in terms of general row-sum coefficients and invokes principal-angle estimates to obtain a positive MPS parent-Hamiltonian gap. The formal proof contains the finite-overlap row reduction for cyclic windows, specializes the coefficients to $c_{ij} = 1/(2(L-1))$, fixes the constant $\gamma_L = 1/(4L)$, and reduces the remaining MPS-specific work to the norm-compression or ordered Friedrichs estimate above.

The Cirac–Perez-Garcia–Schuch–Verstraete 2021 statement is not being challenged. The comparison records that the available formal argument is more explicit than the review paragraph: the exact constants and the cyclic-window blocking convention still have to be compared with the FNW, Nachtergaele, and Kastoryano–Lucia estimates.

Verdict: open gap.

References

[CPGSV21] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product states and projected entangled pair states: Concepts, symmetries, theorems. *Reviews of Modern Physics*, 93(4):045003, 2021.

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[KL18] Michael J. Kastoryano and Angelo Lucia. Divide and conquer method for proving gaps of frustration free spin systems. *Journal of Statistical Mechanics: Theory and Experiment*, 2018(3):033105, 2018.

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